

# Mariam Helal

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## Education

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### University of California, Berkeley

Berkeley, CA, USA

BA in Planetary Science, BA in Astrophysics, Minor in Chemistry

May 2028

- GPA: 4.0/4.0
- Relevant Coursework: Introductory Astrophysics, General Chemistry, Organic Chemistry, Planetary Astrophysics, Calculus, Differential Equations, Linear Algebra, General Physics

### Mission Early College High School

Santa Clara, CA, USA

High School Diploma, AS-T in Physics

May 2024

- GPA: 4.8/4.0
- SAT: 1570

## Research

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### Ramsden Research Fellow

Berkeley, CA, USA

Macdonald Lab

Oct 2024 – present

- Developing protocol for usage of hand-held XRF instrument to analyze and interpret major element compositions of carbonates
- Investigating effects of diagenesis and mineralogy on  $\delta^{13}\text{C}$  values of carbonates recording the Trezona anomaly

### Undergraduate Researcher

Berkeley, CA, USA

Bergner Lab

Feb 2025 – present

- Modeling edge-on protoplanetary disks using radmc3d to analyze relationship between inclination and column density calculations
- Modeling ice emission in protoplanetary disks
- Conducted molecule identification and spectral analysis of ALMA proto-stars to probe phosphorus chemistry in planet-forming regions

### Student Researcher

Fremont, CA, USA

Aspiring Scholars Directed Research Program

Sept 2022 – Dec 2023

- Developed protocol for quantifying microplastic content in California mussels (*Mytilus californianus*) using enzymatic digestion and fluorescence microscopy
- Compiled results into preprint: *Testing a Method for Quantifying Microplastic Content in Mytilus californianus*

## Teaching

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### ASTRON/EPS C162 - Planetary Astrophysics Reader

Berkeley, CA, USA

UC Berkeley Department of Astronomy & Department of Earth & Planetary Sciences

Aug 2025 – present

- Graded student work, including problem sets and exams

### Python DeCal Student Facilitator

Berkeley, CA, USA

UC Berkeley Department of Astronomy

Jan 2025 – present

- Taught lectures on Python programming, plotting, and data analysis techniques
- Co-hosted office hours to provide students with individual support and address coding questions
- Carried out instructional duties including grading homework assignments and mentoring students on final projects

### ASTRON/EPS W12 - Planets Tutor/Reader

Remote

UC Berkeley Department of Astronomy & Department of Earth & Planetary Sciences

June 2025 – Aug 2025

- Interacted with students via discussions and office hours
- Point of contact for sections of the class, assisting student with material-related and administrative questions

## Professional Experience

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### Chief Scientist

NASA L'SPACE Mission Concept Academy

Remote

Jan 2025 – May 2025

- Participated in weekly seminars covering the process of NASA mission concept development and learning from industry professionals
- Led team of 5 through science-delegated tasks, developing a NASA mission concept for a Aerobot system to conduct research on the Venusian atmosphere and surface and navigating surprise roadblocks through reevaluating priorities and descopeing
- Presented final PDR document to board of mentors, providing a summary of our mission and answering clarifying questions

## Extracurriculars

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### Mentor

Undergraduate Laboratory at Berkeley (ULAB), Physics & Astronomy

Berkeley, CA, USA

Sept 2025 – present

- Mentored new undergraduates through research project analyzing ALMA observations of protoplanetary disks

### Telescope Crew Officer

Undergraduate Astronomy Society

Berkeley, CA, USA

May 2025 – present

- Operated & trained telescope crew in usage of Newtonian & Schmidt-Cassegrain telescopes
- Trained in and operated the Richard Treffers 17" PlaneWave telescope, of the UC Berkeley Department of Astronomy
- Co-facilitated weekly Sidewalk Astronomy outreach events to engage the campus community in public observation sessions

### MPS Scholars Mentor

Math & Physical Sciences Department, UC Berkeley

Berkeley, CA, USA

May 2025 – present

- Advised freshman on course selection, research opportunities, and academic planning within the EPS and Planetary Science disciplines
- Hosted regular mentoring meetings to address student questions and provide academic and personal support

### Extrusion Sub-team Member

Student Environmental Resource Center

Berkeley, CA, USA

Sept 2024 – May 2025

- Researched and tested methods for extrusion of recycled 3D printer filament to create a closed-loop campus filament recycling program

### Sensor Placement Co-Lead

Strawberry Creek Monitoring Group

Berkeley, CA, USA

Jan 2025 – Aug 2025

- Created social media posts and graphic design materials for outreach on SCMG activities and creek health
- Co-led sensor placement efforts, finding optimal locations to place sensors monitoring creek activity

## Awards & Honors

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### Ramsden Research Fellowship

Summer 2025, Fall 2025

### Dean's List

Fall 2024

### Presidential Volunteer Service Certificate

Fall 2023

## Skills

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**Lab:** Microscopy, XRF, Mineral Separation

**Coding:** Python, Matplotlib, Pandas, NumPy, SciPy, Seaborn, Cartopy, LaTeX, HTML, CSS, HPC (Savio)

**Programs:** CASSIS, CARTA, radmc3d

**Languages:** English, Arabic